Project Outline

Motivation

PEO grows hard, dense oxide coatings on aluminium via high-voltage electrochemical discharge, giving surfaces with good wear and corrosion resistance. The physics governing oxide growth are poorly understood at the microstructure scale, and existing models rely on empirical fitting that limits their range of applicability.

Goal: build a physics-based computational framework that predicts the microstructural and mechanical properties of PEO coatings on aluminium directly from processing conditions.

Research Roadmap

1
Fracture Model
CP + phase-field fracture
(this poster)

→

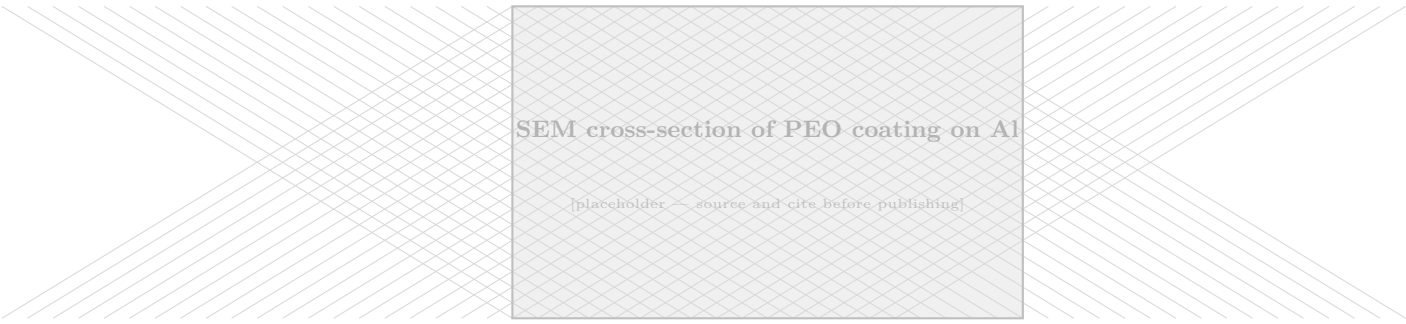
2
Electrolyte–Interface Breakdown Model
high-field ion transport

→

3
Oxide Growth Model
interface kinetics
+ microstructure evolution

→

4
Fully Coupled PEO Growth Model
processing → properties



SEM cross-section of a PEO coating on aluminium showing oxide layer microstructure. Source: [***].

Phase 1 — Current Work (this poster)

`CPFoam`, an Eulerian crystal plasticity framework, is being extended to include phase-field fracture. The crack is treated as an order-parameter field φ_{cr} that evolves under competition between elastic strain energy W_{el} and grain-boundary interfacial energy W_{int}. Results from a two-grain bicrystal under mode I loading show:

- crack nucleation and propagation along the grain boundary
- W_{int} localises at the GB; W_{el} drives nucleation
- the main numerical challenges in the fully coupled regime are identified

Outlook

The aim is an OpenFOAM toolkit that takes processing conditions (voltage, electrolyte, temperature) as inputs and outputs coating microstructure and mechanical properties (hardness, residual stress, crack morphology, oxide thickness) with no empirical fitting.

- **Phase 2:** high-field ion transport and dielectric breakdown at the electrolyte–oxide interface
- **Phase 3:** oxide growth kinetics coupled to the breakdown model, resolving the evolving interface and coating microstructure
- **Phase 4:** fully coupled PEO growth model, validated against experimental coating data on aluminium